

Process Capability Study Procedure

1 Overview

This section describes the basic concept and points to be noted for grasping the process capability and actions for the result in order to clarify that the supplier's process has the ability to consistently achieve the intended quality level.

2 Requirement

2.1 Concept of process capability:

Process capability is the capability related to the quality of a stable process, and the product quality of a standardized process. It is represented by the width of variation " $\bar{X} \pm 3\sigma$ ".

$$\sigma^2 = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

σ : standard deviation

n : number of data

X_i : each data

$\bar{X}$: Average value of each data

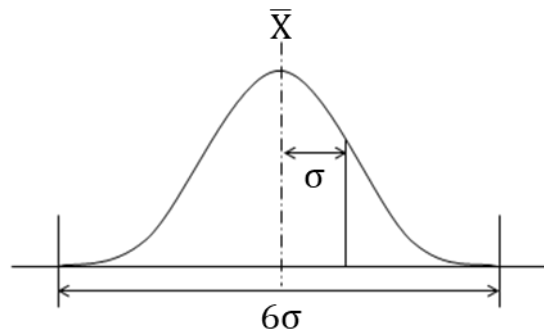


fig 1. Expression of process capability

2.2 Process capability index

The process capability index consists of C_p (an index used when the center of the data distribution (= average value) is in the middle of the upper specification value and the lower specification value),

There is C_{pk} (an index used when the average value deviates from the center of the upper and lower specifications).

The "k" in C_{pk} is derived from the Japanese word "katayori".

2.2.1 Process capability index (C_p)

The 6σ variation range is compared with the product specification range to determine the ability of the process to produce products within the specification.

C_p = standard width / 6σ (in the case of one-sided standard),

$$C_p = \frac{|\text{Standard value} - \text{Average value}|}{3\sigma}$$

1) In case of $C_p \geq 1.00$, the process capability is not sufficient, but almost no defects occur.

(More than 99.7% of the products are within the standard): Perform strict process control and keep it in a controlled state.

If C_p approaches 1, there is a risk of defective products, so take action as necessary.

2) In case of $C_p \geq 1.33$, the process capability is sufficient.

(More than 99.99% of the products are within the standard): Maintain because it is in an ideal state. 1.33, the process capability is sufficient.

2.2.2 Process capability index (C_{pk})

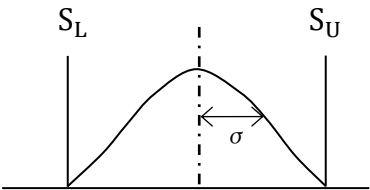
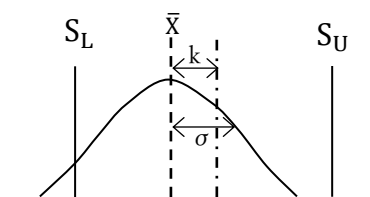
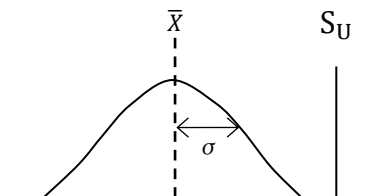
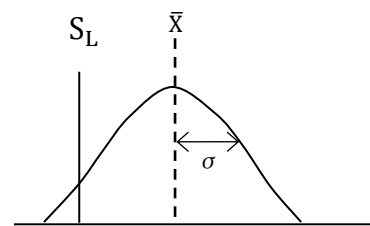
Even if $C_p \geq 1.33$, defects may occur if the average value deviates from the center of the standard.

Determined by the process capability index C_{pk} .

If the average value deviates from the center of the specification, measures should be taken to bring it to the center of the specification.

2.3 Calculation of process capability index

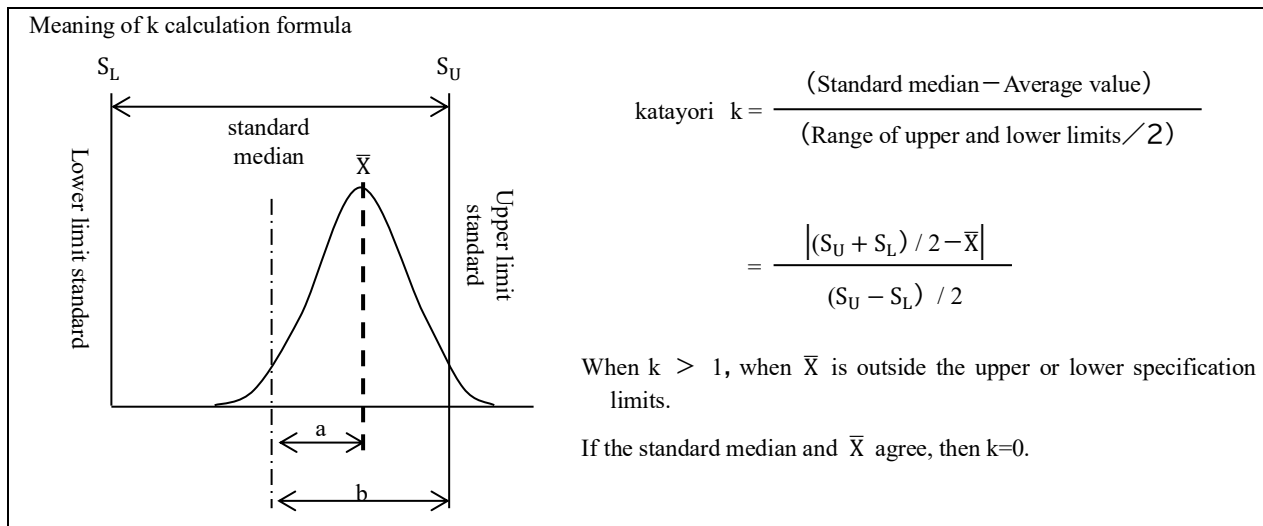
Example of the process capability index calculation is shown below,

Classification		Relationship between distribution and standard	Formula	Calculation example
Both sides standard	C_P		$C_P = \frac{S_U - S_L}{6\sigma}$	$C_P = \frac{52 - 49}{6 \times 0.48}$ $= \frac{3}{2.88} = 1.04$
	C_{PK}		$k = \frac{ (S_U + S_L)/2 - \bar{X} }{(S_U + S_L)/2}$ $C_{PK} = (1 - K) \frac{S_U - S_L}{6\sigma}$ When $k \geq 1$, set $C_{PK} = 0$	$k = \frac{ (52 + 49)/2 - 50 }{(52 - 49)/2}$ $C_{PK} = (1 - 0.3333) \frac{52 - 49}{6 \times 0.48}$ $C_{PK} = 0.69$
One-sided standard	In the case of the upper limit standard (S_U)		$C_P = \frac{S_U - \bar{X}}{3\sigma}$ When $\bar{X} \geq S_U$, $C_P = 0$	$C_P = \frac{52 - 50}{3 \times 0.48}$ $C_P = 1.39$
	In the case of the lower limit standard (S_L)		$C_P = \frac{\bar{X} - S_L}{3\sigma}$ When $\bar{X} \leq S_L$, $C_P = 0$	$C_P = \frac{50 - 49}{3 \times 0.48}$ $C_P = 0.69$

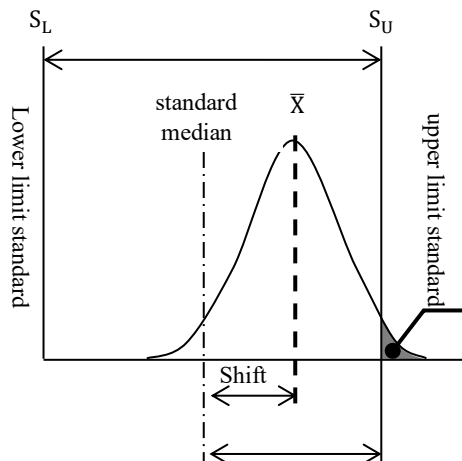
[Example] Average value: $\bar{X} = 50$ standard deviation: $\sigma = 0.48$ upper limit standard: $S_U = 52$

lower limit standard: $S_L = 49$

[Note] k: katayori | |: Absolute value



【Example of worth average】

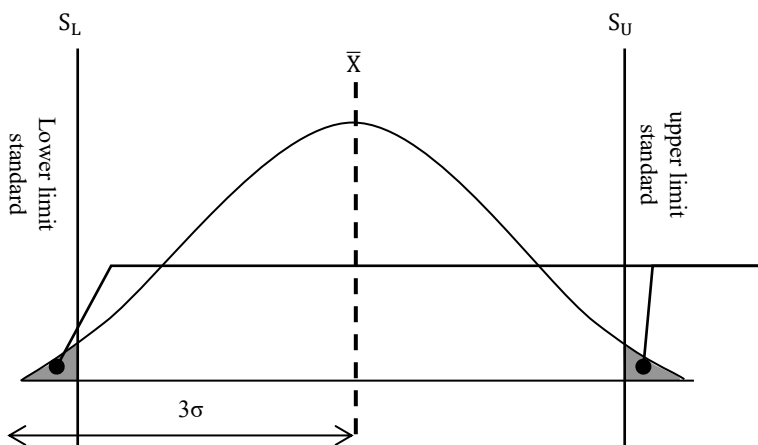


Although the width of the variation is within the product standard value, there is a deviation in the average value, and a bias can be seen.

The bias is large, and there is a possibility that a problem will occur.

Even if C_p is OK, if C_{pk} is NG, There is a possibility of outflow of nonconforming products.

【Example of worth variation】



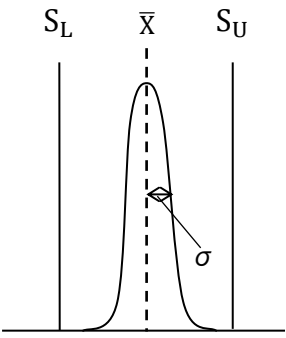
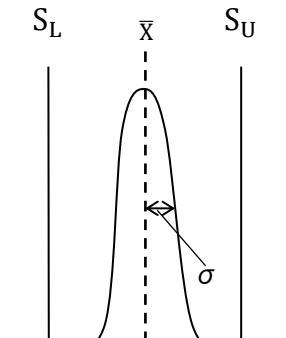
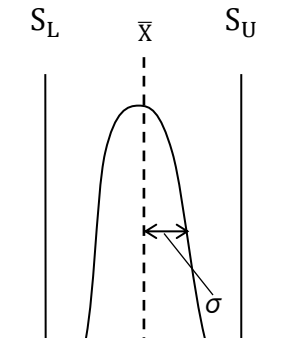
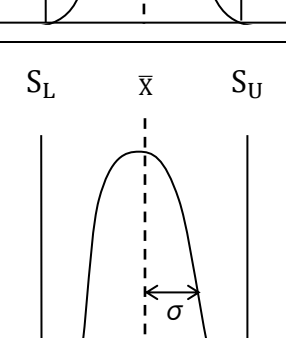
Although the average value is in the middle of the product specification value, the range of variation is large and exceeds the upper and lower specification limits.

There is a high possibility that trouble will occur.

Nonconforming products may occur.

2.4 Process capability judgment and measures

Criteria for judging the presence or absence of process capability based on the process capability index and measures to be taken are as follows.

Relationship between distribution and standard	C_{pk}	Judgement	Measure	Estimated defect rate
	$C_{pk} \geq 1.67$	The process capability is too sufficient.	Inspection method: 100% or sampling inspection No need care if the product variability is slightly greater. Consider ways to simplify management and reduce costs.	0.06/ 100,000pcs or less
	$1.67 > C_{pk} \geq 1.33$	The process capability is sufficient.	Inspection method: 100% or sampling inspection Keep it in ideal condition.	0.06/ 100,000pcs ~ 6.3/ 100,000pcs
	$1.33 > C_{pk} \geq 1.00$	The process capability is not sufficient.	Inspection method: 100% inspection or sampling inspection is performed and a retroactive action procedure is applied to lots with zero nonconformities or potential nonconformances before they are shipped from the supplier. Process improvement is required until $C_{pk} \geq 1.33$. Carry out process control and keep it in a controlled state.	6.3/ 100,000pcs ~ 2.6/ 1000pcs
	$1.00 > C_{pk} \geq 0.67$	Insufficient process capability.	Inspection method: 100% inspection Clarify whether the lack of process capability is due to deviation or variation in the median value, investigate, and implement countermeasures.	2.6/ 1000pcs ~ 45.6/ 1000pcs

2.5 Points to note regarding process capability

2.5.1. Correct sampling: The points for correctly estimating the properties of the population are as follows.

- 1) Random sampling = data under the same conditions are not biased.
- 2) Measurement accuracy = Required to be one digit lower than the standard value.
- 3) Avoidance of contamination of abnormal factors = Data acquisition in a state where factors that should be avoided are not occurring.
- 4) Clarify the definition of lot, and understand lot-to-lot variation. Confirm at least 3 lots or more, and determine whether or not to increase n.

2.5.2. Data Judgment

2.5.2.1 Confirm the distribution and shape of the data. (assuming normal distribution)

If there are multiple cavities, confirm the data for each cavity. In the case of a processing method that has a proven track record, it is also effective to check the width and median value of variation by comparing with the actual results.

If not normally distributed, variable transformation is required

2.5.2.2 Confirm the deviation from the median with $C_p k$.

2.5.2.3 Sufficient n number to judge. If the number of n is small, it is important to make a judgment considering the reliability of the data will drop (increase or correct the number of n if it is barely above 1.33).

2.5.3. Setting parameters with a margin

When considering process capability by judging data, the above paragraphs 3.5.2.1 and 3.5.2.2 are the points to be judged, but if the parameters that affect the results are suppressed, If it can be confirmed that it can be manufactured within the standard even under conditions that deviate slightly from the lower limit or upper and lower limits, it can be said that the process does not deviate from the standard and has process capability. It is important to set conditions with a margin backed up by data.

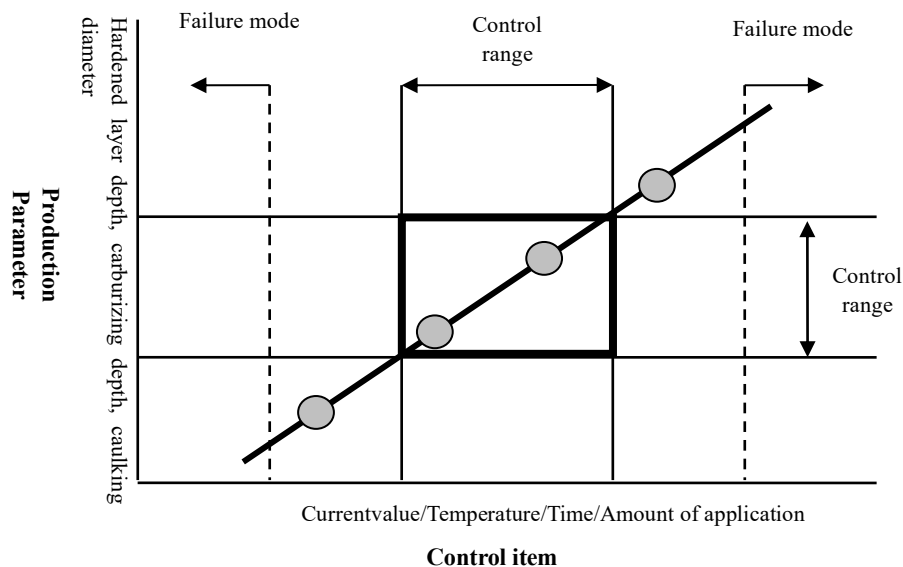


fig 2. Production parameter setting

2.6 Evaluation at the production preparation stage

2.6.1 Concept of 50% tolerance band limit

In the production preparation stage, the number of n is limited, and in order to guarantee the process capability during mass production with that small number of n , the population should aim to come closer to the center of the specification.

If the evaluation of measured values with a small number of n is biased toward the upper and lower limits of the standard, there is a high possibility that the standard will be violated during mass production, and mold correction will be required during mass production.

Therefore, in the production preparation stage, the standard width for evaluation is set at 50% of the drawing-indicated tolerance, and by evaluating the variation, the initial quality is ensured and production preparation activities proceed.

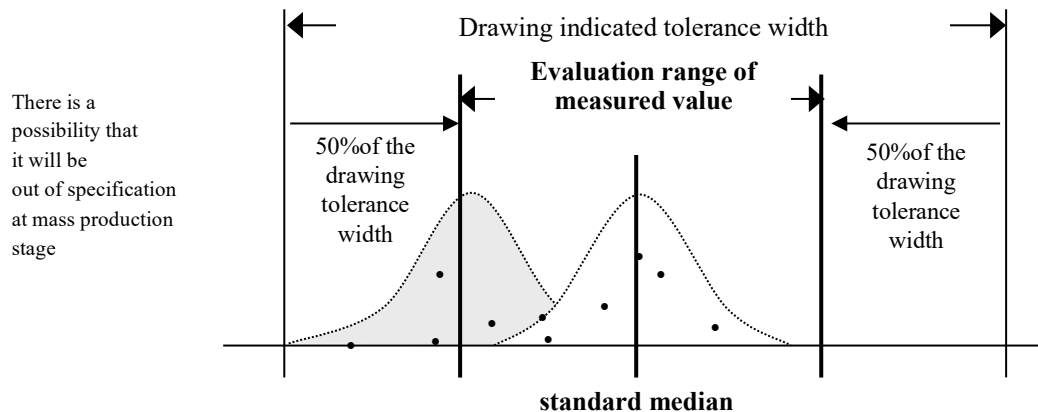


fig 3. 50% tolerance band limit

2.6.2 Judgement Criteria

All measured values shall be within the range of 1/2 of the tolerance width indicated on the drawing.

However, even if the above conditions are not satisfied, it is sufficient if it can be proved that the tolerances are satisfied at the time of mass production.

Note: The above is a way of thinking when the n number in the production preparation stage cannot be secured sufficiently. In the mass production stage, based on the sufficient n number, the process capability is judged by $C_p k$, etc., and necessary improvements are implemented.

The aim of the 50% tolerance band is to set the manufacturing target in the initial stage to the center, rather than reducing the standard width to half, and even if it is out of this range, it is not treated as defective.

2.6.3 Measure for non-standards

2.6.3.1 If there is deviation from the evaluation width of the measured value (50% width of the tolerance width indicated in the drawing) according to the judgment criteria in Section 3.6.2 above, continue the measurement even after the start of mass production. Evaluate the process capability ($C_p k$) based on the data of $n=100$ or more.

